

# Food for Thought: GreenBasket’s approach to helping SNAP Families and Local Farmers towards Self-reliance

Shreyas Patel

sp101@iu.edu

Indiana University, Indianapolis  
Indianapolis, Indiana, United States

Nisarg Rajpura

nrajpura@iu.edu

Indiana University, Indianapolis  
Indianapolis, Indiana, United States

Aditya Nair

nair1@iu.edu

Indiana University, Indianapolis  
Indianapolis, Indiana, United States

Drishti Jogadia

djogadia@iu.edu

Indiana University, Indianapolis  
Indianapolis, Indiana, United States

## ABSTRACT

Indiana grapples with worsening food deserts, underscored by an increase in its food insecurity rate from 9.4% in May 2020 to 13.4% in December 2022, [1] this study investigates the potential of technology to bridge market information gaps and enhance self-sustainability in communities. Existing research focuses on trade and policy reform, and infrastructure investment as interventions to solve food insecurity but overlooks the importance of comprehensive market information dissemination, a gap highlighted by organizations like the FAO and the World Bank. To bridge this gap, our research involved user studies with farmers, food banks, and low-income consumers through surveys, interviews, and low-fidelity evaluations. This approach helped uncover key issues of affordability and market accessibility as underlying barriers to an on-ground solution. We developed Project GreenBasket in response to the aforementioned problems, an online marketplace featuring direct sales channels for farmers, an affordable click-and-collect model for pick-up locations for consumers, and SNAP and nutrition tracking for relevant stakeholders working towards food security. GreenBasket not only addresses the identified gaps in literature but also aims to enhance food security and equity in Indiana, contributing towards the Sustainable Development Goals of Zero Hunger. Our findings reveal the significant potential of technology in improving market access and efficiency, showcasing a scalable model to combat food insecurity.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction design**: *Collaborative and social computing*; • **Applied computing** → Electronic commerce; **Agriculture**.

## KEYWORDS

Food Insecurity, Agricultural Technology, Human-Computer Interaction (HCI), SNAP Benefits Integration, Market Accessibility,

Sustainable Agriculture, Community Self-Reliance, Online Marketplaces, User Experience Design, Food Distribution Logistics

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## 1 INTRODUCTION

We met with the members of Circles Indy at Tabernacle Presbyterian Church for their monthly meetup of families under the Supplementary Nutritional Assistance Program (SNAP) [7], more commonly known as food stamps in the United States. This program aims to provide nutritional support to eligible low-income individuals and families. SNAP benefits are used to buy eligible food in authorized retail food stores. One-fifth of them live in food deserts, 30% live in households without cars, and more than 40% have to walk more than half a mile carrying groceries [14]. The average family of four on SNAP has a net monthly income of \$2,313, (11% below the national poverty line) with \$111.6 going towards food for each member [6][15]. In collaboration with Agrinovus Indiana for the HungerTech Innovation Challenge, we worked with Gleaners (the largest food distribution center in Indiana), Purdue Center for Food Demand Analysis and Sustainability, and Indiana Farm Bureau to understand the operational complexities in providing weekly groceries to SNAP families, working on a broader challenge in Indiana, where food insecurity has risen, as shown by the increase in the state’s food insecurity rate from 9.4% in May 2020 to 13.4% in December 2022 [1]. They face limited supermarket options, compounded by high fuel costs and delivery charges, making access to nutritious food a significant challenge. The Food and Agriculture Organization (FAO) of the United Nations identifies three primary intervention strategies to combat food insecurity: trade and policy reform, infrastructure investment, and efficient market information dissemination [5]. Extensive progress has been made in creating frameworks for planning and designing interventions in Trade and Policy Reform [3]. Several organizations, including the United States Department of Agriculture (USDA), World Health Organisation (WHO), and the World Bank, are working toward the second objective of Infrastructure Investment improvement [12]

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[16]. A glaring gap remains in frameworks and research toward the provision and utilization of efficient market information to help stakeholders in the food chain provide for people in need [13][2]. This gap critically affects all levels of food insecurity, from the local household [11] to the volatility of food prices [8][9], exacerbated climate change [4], and the inability to detect early onset famine [10]. Despite recognition of this issue by global organizations such as the USDA, WHO, and the World Bank, there exists a notable deficiency in frameworks to efficiently relay market data to stakeholders within local food ecosystems [12] [16].

Through this paper, we investigate technology's role in fostering self-sustainable communities and empowering Indiana's local farmers to support vulnerable groups. We propose GreenBasket - a self-reliant, technology-driven solution: an online marketplace tailored to the specific needs of our stakeholders. This marketplace is designed to offer real-time market analytics, enable direct transactions between farmers and consumers, and seamlessly integrate SNAP benefits. Features like CropSpots for easy produce pick-up and educational initiatives to enhance SNAP benefit utilization are also included. Our research extends into the anticipated impacts of GreenBasket in mitigating food insecurity in Indiana, using data from our user research to address critical barriers to healthy eating among low-income residents, emphasizing affordability and accessibility. This intends to contribute to achieving the sustainable development goal of Zero Hunger, enhancing food security, improving nutrition, and promoting sustainable agricultural practices.

## 2 USER RESEARCH

### 2.1 Objective and Methodology

We focused on two main stakeholders for user research: farmers and low-income consumers, aiming to identify barriers they face in addressing food security in their communities. To collect data, we used demographic surveys, generative interviews, regular interactions with stakeholders from food banks, and follow-up surveys for the problem and feature prioritization. Our methodology for participant recruitment spanned Facebook Groups, the Purdue Agricultural Extension Service, and local Indiana community networks. Eligible participants, verified through screening questions, were contacted via email for the exploratory study. Participants, including 20 farmers, 12 low-income individuals, and 18 industry professionals from food distribution centers, received a \$30 Amazon gift card post-session. Farmers were categorized as small-scale (less than 50 acres, annual revenue under \$50,000, fewer than 5 workers) or medium-scale (50 to 200 acres, annual revenue \$50,000 to \$250,000, 5 to 20 workers). Food distribution centers had to be registered nonprofits in Indiana and operational for at least two years. Low-income consumers met the criteria of being below the poverty line and having experience with SNAP. A subset of 12 farmers and 8 low-income individuals participated in a follow-up survey to prioritize problems.

### 2.2 Data Analysis

In our study, we conducted semi-structured interviews with low-income individuals focusing on food access challenges and with farmers on barriers to providing food to communities. A second round of interviews with industry experts delved into constraints in

implementing food security solutions. Inductive coding from these interviews identified key challenges: farmers cited three themes Market Accessibility Issues, Inefficiencies in Farm Operations, and Lack of Real-Time Market and Crop Data. Low-income participants mentioned Difficulty Accessing Affordable Fresh Produce, Transportation and Accessibility Challenges, and Limited Knowledge of SNAP Benefits. Two surveys were conducted: an initial demographic survey and a second to rank identified problems. Among farmers, 65% managed less than 50 acres, and 35% between 50 and 200 acres. Most were between the ages of 35 and 55 and had limited digital marketplace experience. Low-income participants, mainly aged between 25-45 and predominantly from households below Indiana's poverty line, reported high smartphone and internet usage but limited access to fresh produce. In the follow-up survey, participants ranked problems based on thematic codes, with the option to provide open-ended feedback. The ranking averaged across participants helped create a problem description based on thematic analysis and responses. This ranking process, conducted on a scale of 3, prioritized the key problems, as demonstrated in table 1 and table 2.

### 2.3 Feature Identification and Analysis

Ten initial features were identified over two brainstorming sessions between researchers and one round of interviews with industry stakeholders to eliminate ideas with barriers to practical implementation. Three features were theorized to address the three problems for farmers - Market Analytics, Marketplace, and MyFarm Farmer Management Tool. Seven features were created addressing the three problems coded for low-income participants: CropSpots, Group Deliveries and Cost Sharing, SNAP/ Electronic Benefits Transfer (EBT), Recipes, Donation Channel, and SNAP Outreach. We evaluated each feature's feasibility and potential impact on farmers and consumers.

## 3 DESIGN PROCESS

### 3.1 Design Ideation, Wireframe Development and User Feedback Sessions

Before diving into the design process, we sought inspiration from existing solutions, web applications, and mobile apps that had similar functionalities or were related to the features identified for our solution. To consolidate the ideas and inspirations gathered, we created a moodboard to capture various interactions, flows, and design elements from the researched sources. It served as a visual reference and inspiration guide for the design team, enabling them to visualize potential design directions and user experience strategies. With the moodboard as a reference, we began brainstorming and ideating how best to integrate the identified features into a cohesive, user-friendly interface. This step involved sketching initial concepts, discussing possible layouts, and determining how each feature would be represented within the application. Based on the ideation phase, we developed 10 low-fidelity wireframes addressing each of the features from table 1 and table 2. These wireframes focused on creating intuitive and efficient screens that would guide users through the various features of GreenBasket. 6 farmers and 4 low-income participants from the earlier stage of the study agreed to a follow-up user feedback study. Participants were presented with low-fidelity wireframes of the proposed Minimum

| Key Problems                      | Average Rank | Problem Description                                                                                                                       | Suggested Features                                                                                                       |
|-----------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Market Accessibility              | 1.95         | Farmers expressed a significant need for better channels to reach consumers directly and market their produce more effectively.           | <b>Marketplace::</b> Enables direct sales to consumers, improving market access.                                         |
| Lack of Real-Time Market Data     | 2            | Farmers reported a gap in accessing timely market data and crop health information, which is crucial for informed agricultural practices. | <b>Market Analytics:</b> Delivers real-time data on market trends and product performance, aiding in strategic planning. |
| Inefficiencies in Farm Operations | 2.05         | The feedback highlighted challenges in managing farm resources efficiently, with a need for streamlined operations and decision-making    | <b>MyFarm Farmer Management Tool:</b> Offers comprehensive management of farm operations, enhancing efficiency.          |

Table 1: Prioritized Key Problems and solutions for farmers (n=20)

| Key Problems                                       | Average Rank | Problem Description                                                                                                                  | Suggested Features                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Affordability of Fresh Produce                     | 1.83         | Participants expressed concerns over the high cost and inaccessibility of fresh, healthy food options within their budget.           | <b>EBT Card Payment:</b> Facilitates using SNAP benefits for purchasing affordable produce.<br><br><b>Food Donation Channel:</b> A dedicated section on the platform for individuals and organizations to donate food.                                                                                                                                                               |
| Transportation and Accessibility Challenges        | 1.92         | There was a significant emphasis on the difficulty of physically reaching food distribution points, especially in low-density areas. | <b>CropSpots:</b> Provides convenient pick-up locations to access fresh produce easily.<br><br><b>Group Deliveries and Cost Sharing:</b> Addresses cost concerns by enabling shared delivery expenses.                                                                                                                                                                               |
| Limited Knowledge and Utilization of SNAP Benefits | 2.17         | Feedback revealed a lack of awareness and understanding of SNAP benefits, hindering their effective utilization.                     | <b>SNAP Outreach:</b> Educates on the benefits and process of SNAP, enhancing utilization.<br><br><b>Fresh Recipes:</b> Encourages healthy cooking practices by offering recipes based on purchased produce.<br><br><b>Food Security and Health Analytics:</b> Offers insights into food security impact and nutritional information, with interactive reports for healthier eating. |

Table 2: Prioritized Key Problems and solutions for Low-Income Participants (n=12)

Viable Product (MVP) features. Participants were led through a usability task from each of the sketches Fig. 1.

3.2 Prioritization of Features

Feedback was collected qualitatively regarding feature usability and the participants were then asked to rank the three features for farmers and six features for low-income participants based on impact and ease of use Fig. 2.

3.3 Incorporating Feedback into Design

Based on user preferences and requirements; we adjusted the design, modifying and even removing certain features. The wireframes were updated iteratively, integrating changes and undergoing retesting with users to ensure a refined, user-centered design. Utilizing a prioritization map, we carefully considered user feedback and prioritized features that aligned with user needs and the project’s goals. This process led us to finalize a set of features for the platform: a Marketplace with CropSpots, Group Deliveries, and EBT Payment, along with a dedicated dashboard for farmers featuring market analytics. Additionally, we included a food security and

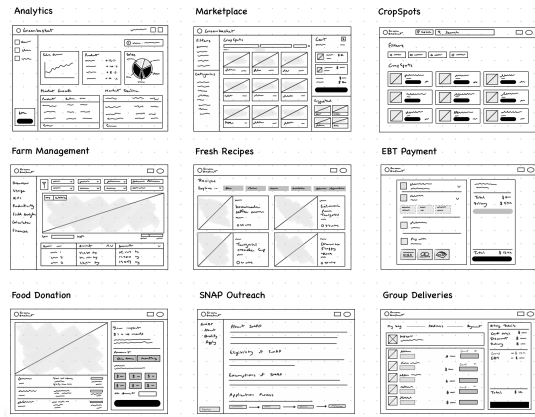


Figure 1: Sketches of initial concepts



Figure 2: Feature prioritization

health analytics feature for CropSpots, such as food banks, hubs, and pantries. This feature was designed in response to the need of food distribution centers to track SNAP payments and nutritional impact of the centers served.

### 3.4 GreenBasket - Towards zero hunger

From the product, the consumer and farmer flows, as well as the impact on CropSpots like food banks, hubs, and pantries, are outlined as follows:

#### 3.4.1 Consumer User Flow:

- (1) Marketplace and Cropspots (Fig. 3): Consumers can easily order a wide range of fresh produce through the marketplace, which offers comprehensive produce categories and convenient navigation and search refinement features. The click-and-collect model allows consumers to pick up their orders from centralized CropSpots locations in their area. CropSpots display the assortment of fresh produce available and upcoming, with options to filter products by SNAP eligibility. Consumers can view the items available at their

nearest CropSpot, enhancing the convenience and accessibility of fresh food.

- (2) Group Deliveries and EBT Payment (Fig. 4): Consumers can see how many people are ordering the same item and can choose to split the delivery costs among themselves for additional savings. The platform offers a selection of preferred payment modes, including EBT payments, enabling consumers to utilize their SNAP benefits efficiently. This feature makes it easier for consumers to manage their food expenses while accessing various food options.

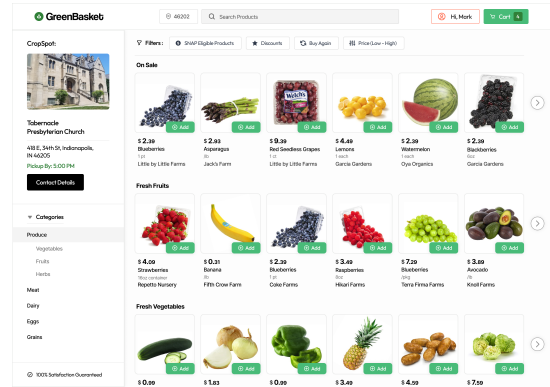


Figure 3: Marketplace and Cropspots

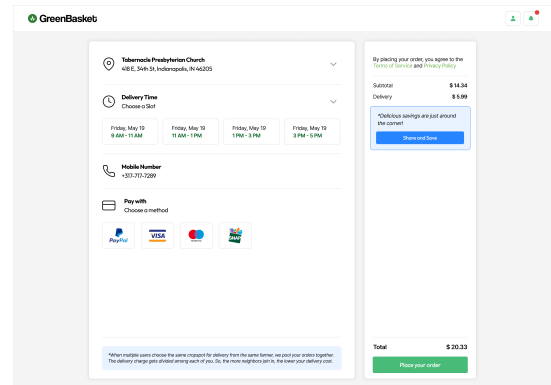


Figure 4: Group Deliveries and EBT Payment

#### 3.4.2 Farmer User Flow:

- (1) Marketplace and Dashboard (Fig. 5): Farmers benefit from direct sales to consumers, achieving better profit margins. The platform includes tools for tracking and managing orders, including those from SNAP EBT users, facilitating direct customer relationships. Farmers have access to comprehensive data on monthly sales, individual product performance, and a detailed analysis of product sales, helping them to better understand and cater to market demands.
- (2) Market Analytics (Fig. 6): Farmers can manage their profiles and keep track of farm analytics customized to their

products. This includes access to data on total monthly sales, individual product performance, and sales breakdowns. Real-time alerts on price updates and the ability to study market growth and decline patterns for specific products are provided. The platform offers a comprehensive understanding of market trends, enabling farmers to make informed decisions, navigate the agricultural industry dynamically, and analyze projected sales and Return on Investment (ROI).

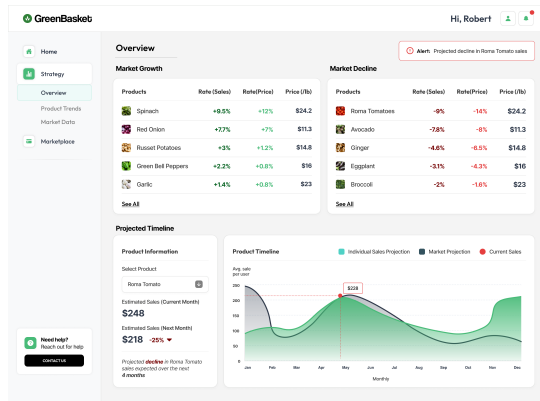


Figure 5: Dashboard

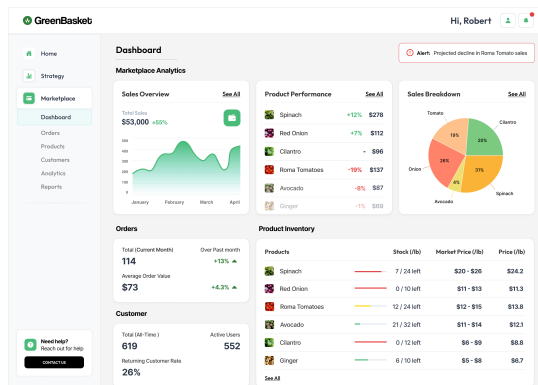


Figure 6: Market Analytics

### 3.4.3 For food distribution centers (Food Banks, Hubs, Pantries, etc.)

- (1) Food Security and Health Analytics: CropSpots can keep track of the impact made towards solving food insecurity and understand the nutritional value of the food consumed to improve health outcomes. This feature allows CropSpots to monitor and enhance their contribution to community well-being, ensuring that they are effectively addressing local food security and health needs.

The high-fidelity mockups visually represent these key features, providing a clear and engaging depiction of how each component of the GreenBasket platform functions and benefits its users. These mockups are crucial for stakeholders to visualize the user experience and understand the practical application of each feature in the MVP.

## 4 EVALUATION AND FUTURE WORK

To evaluate our solution, we chose to test it at farmers’ markets because it serves as an accessible in-person touchpoint, allowing us to directly observe the entire supply chain from farmers to consumers. This setting allowed us to collect detailed feedback offering a holistic view of market dynamics. We are working with Growing Places Indy, a volunteer organization that manages local urban farms, organizes local farmers’ markets, and triple matches SNAP dollars, providing \$2 for fresh fruits and vegetables for every \$1 of SNAP dollars spent by low-income neighbors at a farmers’ market. Two tests were conducted at the Indy City Farmers Market and the Winter Farmer Market at 16Tech, with three farmer market managers, five farmers, and eight farmer market participants. Three tasks were tested; first, a farmer market manager viewed the impact made at their farmer market through a CropSpot dashboard to measure the SNAP payments and Nutritional impact made for deliveries to their farmer market. Second, a farmer views the farmer dashboard to view a breakdown of their sales to understand market demand and then views the market analytics dashboard to make decisions on produce to grow and sell for future seasons. Third, a consumer was tasked with selecting a location for delivery, selecting produce farmers are willing to deliver to the location, and paying through a SNAP EBT card. All three stakeholders were asked for feedback on barriers and additions required to a real-life solution. The feedback was inductively coded, resulting in the following themes.

Farmer Market Managers highlighted the need for integrated vendor management tools to streamline the organization of applications from various farmers, a more efficient digital platform for managing SNAP transactions, and features for generating comprehensive impact reports for securing government funding. Farmers voiced the need for Point of Sales (POS) integration to manage inventory across different sales channels effectively and a simplified account setup process for applying to farmers’ markets. Finally, consumers expressed a preference for regular delivery options for specific produce and suggested the integration of food bank inventories into GreenBasket to enhance food accessibility.

## 5 CONCLUSION

We intend to implement the first version of GreenBasket at the Winter Farmer’s Market at 16 Tech Innovation District in an attempt to understand its real-world impact. With the participation of 65 farmers, we hope our study can inform similar efforts in other communities. The goal is to share insights and learnings to improve food security and self-sufficiency collectively. Our solution aims to meet the immediate needs of SNAP families and also supports local farmers in effectively reaching vulnerable communities. The platform’s design aims for a sustainable model that can be implemented independently of external funding and intervention. There are certain limitations we recognize, namely the need to address challenges in transportation, inventory management, and awareness. With the publication of this literature, we hope to invite more feedback and diverse implementations of our solution to help bridge the aforementioned gaps. As a small step towards achieving Sustainable Development Goal 2 (Zero Hunger), we aim to encourage collective action to create meaningful change in the fight against food insecurity.

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## REFERENCES

- [1] Raquel Bahamonde. 2023. Tackling and Solving Food Insecurity with Private Sector Innovations. <https://www.cicpindiana.com/agrinovusindianasolvingfoodinsecurity/>
- [2] Lynn Dombrowski, Jed R. Brubaker, Sen H. Hirano, Melissa Mazmanian, and Gillian R. Hayes. 2013. It takes a network to get dinner: designing location-based systems to address local food needs. In *Proceedings of the 2013 ACM international joint conference on Pervasive and ubiquitous computing (UbiComp '13)*. Association for Computing Machinery, New York, NY, USA, 519–528. <https://doi.org/10.1145/2493432.2493493>
- [3] Azam Doustmohammadian, Fatemeh Mohammadi-Nasrabadi, Nastaran Keshavarz-Mohammadi, Melika Hajjar, Sepideh Alibeyk, and Maryam Hajigholam-Saryazdi. 2022. Community-based participatory interventions to improve food security: A systematic review. *Frontiers in Nutrition* 9 (2022), 4. <https://www.frontiersin.org/articles/10.3389/fnut.2022.1028394>
- [4] FAO. 2015. *Climate change and food security: risks and responses*. FAO, Rome, Italy. <https://www.fao.org/documents/card/en?details=82129a98-8338-45e5-a2cd-8eda4184550f>
- [5] FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS. 2003. Chapter 5. Assessing the impact of trade reforms on food security[62]. <https://www.fao.org/3/y4671e/y4671e09.htm>
- [6] Food and Nutrition Service, U.S Department of Agriculture. 2023. SNAP Eligibility | Food and Nutrition Service. <https://www.fns.usda.gov/snap/recipient/eligibility>
- [7] Food and Nutrition Service, U.S Department of Agriculture. 2024. Supplemental Nutrition Assistance Program (SNAP) | Food and Nutrition Service. <https://www.fns.usda.gov/snap/supplemental-nutrition-assistance-program>
- [8] Matthias Kalkuhl, Joachim Von Braun, and Maximo Torero (Eds.). 2016. *Food Price Volatility and Its Implications for Food Security and Policy*. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-319-28201-5>
- [9] Lisa Klug and Wolfgang Prinz. 2023. Fair prices for sustainability in agriculture and food. Requirements and design options for a data-based transparency system.. In *Proceedings of the 24th Annual International Conference on Digital Government Research (DGO '23)*. Association for Computing Machinery, New York, NY, USA, 496–507. <https://doi.org/10.1145/3598469.3598525>
- [10] P. Krishna Krishnamurthy, Richard J. Choularton, and Peter Kareiva. 2020. Dealing with uncertainty in famine predictions: How complex events affect food security early warning skill in the Greater Horn of Africa. *Global Food Security* 26 (Sept. 2020), 100374. <https://doi.org/10.1016/j.gfs.2020.100374>
- [11] Mahwish Munawar, Xu Shiwei, Yu Wen, and Muhammad Luqman. 2021. Investigating Relationship of Food Security with Market Approachability with respect to Household Food Insecurity Access Index. *Journal of Economic Impact* 3, 3 (Dec. 2021), 130–136. <https://doi.org/10.52223/jei3032101>
- [12] NIFA. 2022. Community Foods Projects Improving National Food Security | NIFA. <https://www.nifa.usda.gov/about-nifa/blogs/community-foods-projects-improving-national-food-security>
- [13] Juliet Norton, Birgit Penzenstadler, and Bill Tomlinson. 2019. Implications of Grassroots Sustainable Agriculture Community Values on the Design of Information Systems. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (Nov. 2019), 34:1–34:22. <https://doi.org/10.1145/3359136>
- [14] Unai Miguel Andres and Ross Tepe. 2019. Getting Groceries: Food Access Across Groups, Neighborhoods, and Time. [https://www.savi.org/feature\\_report/getting-groceries-food-access-across-groups-neighborhoods-and-time/](https://www.savi.org/feature_report/getting-groceries-food-access-across-groups-neighborhoods-and-time/)
- [15] U.S. Department of Health & Human Services. 2024. Poverty Guidelines. <https://aspe.hhs.gov/topics/poverty-economic-mobility/poverty-guidelines>
- [16] World Bank. 2023. What is Food Security? There are Four Dimensions. <https://www.worldbank.org/en/topic/agriculture/brief/food-security-update/what-is-food-security>